REGINA ENGINEERING COMPETITION 2026

Autonomous Dispatch Planner

A Resilient Graph-Based Emergency Response Engine for 90 Simulated Days

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CONCEPTUAL & STRATEGY
Tandia

ARCHITECTURE & CODE
Abraham

The Emergency Response Challenge

Managing a severe Saskatchewan weather crisis under extreme operational constraints.

Complete Uncertainty

The engine never reveals the future. Incidents, road closures, and vehicle breakdowns appear only when they occur.

Strict Deadlines

Late arrivals forfeit lives entirely and keep vehicles committed to wasted trips across the regional network.

Chronic Care Mandate

The regional care home demands physical visits every 24 hours to prevent permanent resident loss.

System Architecture & HTTP Protocol

Clean, dependency-light Node.js architecture communicating seamlessly with the local simulation engine.

Engine & Server Layer

GET /state: Pulls real-time topological snapshot.

POST /dispatch & /move: Executes fleet commands.

POST /advance: Steps time forward to pending triggers.

Robust Error Handling

Try/Catch Wrappers: Intercepts HTTP 4xx rejections.

Zero Stalling: Failed actions never crash the core loop.

Clean Resets: Fully supports scenario reloads instantly.

Evolution & Version History

Iterative development from a basic sequential stub to a global heuristic powerhouse.

Iteration 1

The Stub Loop

Connected HTTP client to engine, advancing time with zero decisions (Score: 0).

Iteration 2

Greedy Dispatch

Assigned first available car to first open emergency. Prone to severe inefficiencies.

Final Version

Global Heuristic

Dijkstra pathfinding + Global board pairing + Smart Care Home memory (Score: 90%).

Design Rationale: Why Our Architecture Won

Defending key architectural choices against rigorous problem constraints.

Dijkstra's Optimality

Accurately handles restricted roads (1.5x weights) and closed road bypasses dynamically without spatial heuristic requirements.

Global Pairing over Greedy

Evaluating all car-emergency permutations simultaneously prevents slow vehicles from stealing high-value missions from fast units.

Graceful Triage

Intentionally cutting losses on unreachable nodes ensures maximum preservation across viable communities.

Code Breakdown & Team Division

Modular separation of concerns powering the dispatch application.

Conceptual & Strategy

Led by Tandia: Focused on rule interpretation, efficiency ratios, risk trade-offs, and chronic care protection thresholds.

Architecture & Implementation

Led by Abraham: Built the `EngineClient` HTTP wrapper, Dijkstra graph search, and the global heuristic sorting matrix.

Performance & Live Dashboard

Verifying end-to-end operational success across training and judging scenarios.

90-DAY JUDGING SCENARIO RESULT

211 / 233

~90% Lives Saved (Optimal Limit Reached)

Interface & Usability

- Real-time Express & Tailwind Web Dashboard
- Color-coded fleet status and incident tracking
- Fully compliant with all competition deliverables

Thank You

Ready for live demonstration, code diff verification, and technical Q&A.

Node.js + Express

Dijkstra Graph Routing